Lecture 4 (More Modeling Examples)

January 24, 2026

We will give a few more modeling examples. These will cover a few common tricks for transforming LPs and some of the relevant tricks that often appear when modeling.

Ex 1)

Fire Station Scheduling

In the town of Pullman, there are 50 trained firefighters. The firefighters are required to spend 4 days straight at the station followed by 3 days off. The number of firefighters needed for the ith day of the week is given by d_i. The fire station wishes to schedule the required number of firefighters for each day of the week while minimizing the number of total of firefighters hired. Model this scenario as an LP.

Model

Let N_i be the number of firefighters scheduled starting on day i of the week (for example, N_2 will be the number of firefighters who will work the Monday - Thursday shift). Then we see

$$\text{minimize } \sum_{i=1}^{7} N_i$$

as the objective function, and

$$N_1 + N_7 + N_6 + N_5 \geq d_1$$

$$N_2 + N_1 + N_7 + N_6 \geq d_2$$

$$N_3 + N_2 + N_1 + N_7 \geq d_3$$

$$N_4 + N_3 + N_2 + N_1 \geq d_4$$

$$N_5 + N_4 + N_3 + N_2 \geq d_5$$

$$N_6 + N_5 + N_4 + N_3 \geq d_6$$

$$N_7 + N_6 + N_5 + N_4 \geq d_7$$

as the minimum staffing constraints

$$\sum_{i=1}^{7} N_i \leq 50$$

as the max number of employee constraint

$$N_1, N_2, \ldots, N_7 \geq 0$$

as the sign constraint. We can convert this to standard form as shown below:

$$\text{minimize } \sum_{i=1}^{7} N_i$$

as the objective function, and

$$N_1 + N_7 + N_6 + N_5 - s_1 = d_1$$

$$N_2 + N_1 + N_7 + N_6 - s_2 = d_2$$

$$N_3 + N_2 + N_1 + N_7 - s_3 = d_3$$

$$N_4 + N_3 + N_2 + N_1 - s_4 = d_4$$

$$N_5 + N_4 + N_3 + N_2 - s_5 = d_5$$

$$N_6 + N_5 + N_4 + N_3 - s_6 = d_6$$

$$N_7 + N_6 + N_5 + N_4 - s_7 = d_7$$

as the minimum staffing constraints

$$\sum_{i=1}^{7} N_i + s_8 = 50$$

as the max number of employee constraint

$$N_1, N_2, \ldots, N_7 \geq 0$$

$$s_1, s_2, \ldots, s_8 \geq 0$$

Ex 2)

Linear Classification

Suppose we have a set of N data points of the form $D_i = (\vec{a}_i, b_i)$. The vector $\vec{a}_i \in \mathbb{R}^n$ is called the feature vector. This corresponds to the known information about the data point. The value b_i corresponds to the classification of the data point D_i. We wish to find a vector $\vec{x}$ (called the parameter vector), such that b_i is close to $\vec{a}_i \cdot \vec{x}$. This closeness is found by taking the residual:

$$r_i = \left| \vec{b}_i - \vec{a}_i \cdot \vec{x} \right|$$

When creating the classification vector, we wish to minimize the residual error. We can model this classification as an LP using a few different possible formulations:

Model 1 (Residual Sum)

Suppose we wish to minimize the sum of residuals. That is:

$$\text{minimize } \sum_{i=1}^{N} r_i$$

Each residual $r_i = \left| \vec{b}_i - \vec{a}_i \cdot \vec{x} \right|$. Even though the absolute value is non-linear, we can break this into two linear constraints. Notice that

$$\vec{b}_i - \vec{a}_i \cdot \vec{x} \leq r_i$$

$$-\left(\vec{b}_i - \vec{a}_i \cdot \vec{x}\right) \leq r_i$$

as one of the two left sides will be negative. The other will be equal to r_i. This allows the following LP formulation (in order to replace the absolute value with two constraints).

$$\text{minimize } \sum_{i=1}^{N} r_i$$

subject to the absolute value constraints:

$$\vec{b}_i - \vec{a}_i \cdot \vec{x} \leq r_i, \ \forall i$$

$$-\left(\vec{b}_i - \vec{a}_i \cdot \vec{x}\right) \leq r_i, \ \forall i$$

and the sign constraints:

$$r_i \geq 0, \ \forall i$$

$$\vec{x} \in \mathbb{R}^n$$

If we wish to get this into standard form, then each entry of $\vec{x}$ needs to be non-negative. This means we can add a few more constraints to achieve this:

$$\text{minimize } \sum_{i=1}^{N} r_i$$

subject to the absolute value constraints:

$$\vec{b}_i - \vec{a}_i \cdot \vec{x} + s_{i1} = r_i, \ \forall i$$

$$-\left(\vec{b}_i - \vec{a}_i \cdot \vec{x}\right) + s_{i2} = r_i, \ \forall i$$

the positivity constraint for $\vec{x}$

$$\vec{x} = \vec{x}^+ - \vec{x}^-$$

and the sign constraints:

$$r_i \geq 0, \ \forall i$$

$$s_{i1}, s_{12} \geq 0$$

$$\vec{x}^+, \vec{x}^- \geq 0$$

Model 2 (Max Residual)

Suppose we wish to minimize the maximum residual. That is:

$$\text{minimize } \max_i r_i$$

This can be modeled as an LP with clever inequalities. We can use the same definition of r_i, but instead of creating a separate variable for each i that must be positive, we merely create one variable. This yields the LP:

$$\text{minimize } z$$

s.t.

$$\vec{b}_i - \vec{a}_i \cdot \vec{x} \leq z, \ \forall i$$

$$-\left(\vec{b}_i - \vec{a}_i \cdot \vec{x}\right) \leq z, \ \forall i$$

with the sign constraints

$$z \geq 0$$

$$\vec{x} \in \mathbb{R}^n$$

Or in standard form:

$$\text{minimize } z$$

s.t.

$$\vec{b}_i - \vec{a}_i \cdot \vec{x} + s_{i1} = z, \ \forall i$$

$$-\left(\vec{b}_i - \vec{a}_i \cdot \vec{x}\right) + s_{i2} = z, \ \forall i$$

$$\vec{x} = \vec{x}^+ - \vec{x}^-$$

with the sign constraints

$$z \geq 0$$

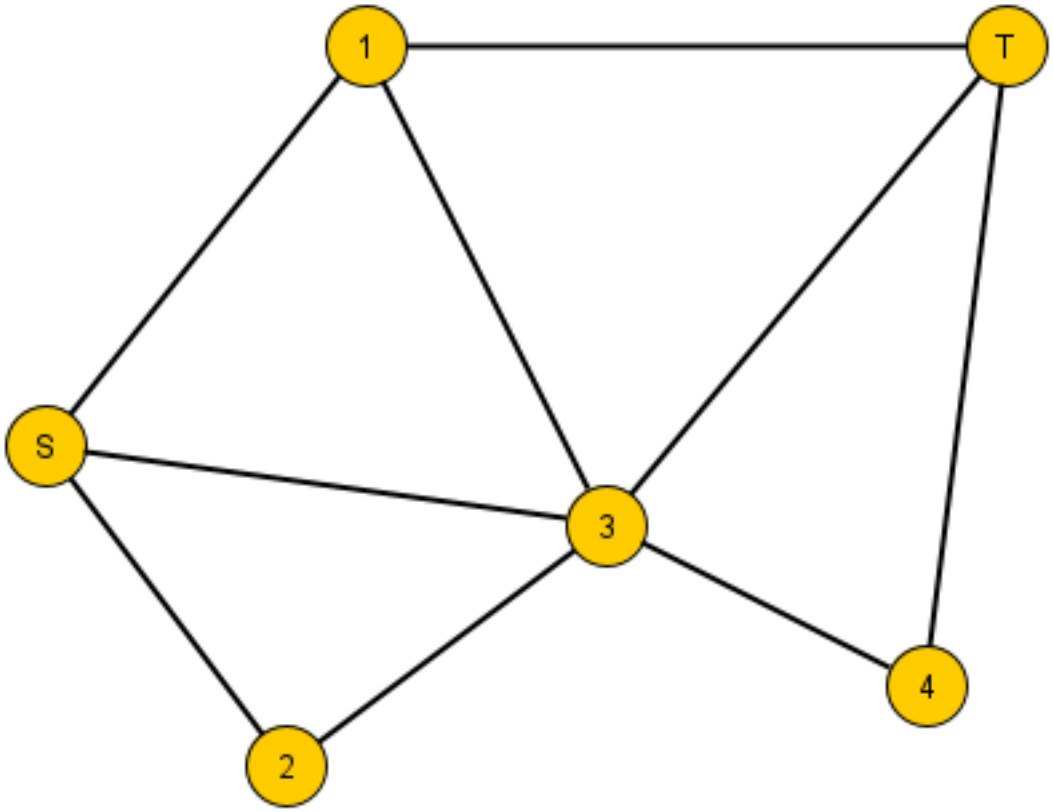
$$s_{i1}, s_{12} \geq 0$$

$$\vec{x}^+, \vec{x}^- \geq 0$$

Ex 3)

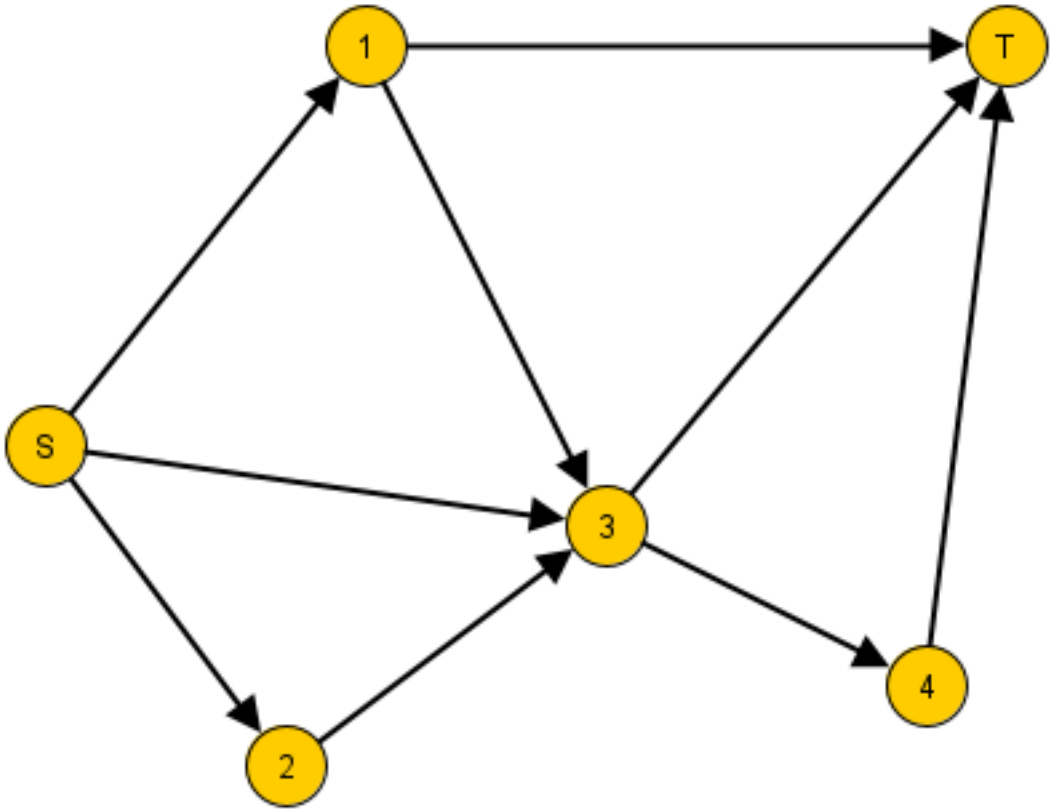
Network Flow

A common problem in linear programming involves efficiently moving goods through a network. This may be routing goods through a network of streets. This may be packets of information flowing through a computer system. This could be efficient communication between different people. There is an entire class at WSU on these types of problems (typically offered every other Fall). The network flow formulation tends to yield even more efficient algorithms, however it can still be modeled more generally as an LP. Consider the following example below. A factory (labeled S in the diagram) wishes to provide N units of goods to the store (labeled T in the diagram). There are a collection of railroad depots along the route (each given a number). The depots are connected with tracks with a capacity of u_{ij} goods between depots i and j. Each track is allowed to have a train go in either direction. Each track has a cost of c_{ij} per good shipped along it. Find the plan that minimizes the cost of shipping N goods from S to T while obeying capacities of the tracks.



Model

First, we will assign arrows along each track. The number of goods shipped along the direction of the arrow from track from i to j will be denoted x_{ij}. If goods are shipped in the direction opposite of the arrow, then we will have a negative value for x_{ij}.



Let S be the collection of all ordered pairs (i, j) such that there is a track with an arrow pointing from i to j. This yields the following formulation:

$$\text{minimize} \sum_{i,j \in S} c_{ij} x_{ij}$$

s.t.

$$x_{s1} + x_{s2} + x_{s3} = 50$$

$$x_{1t} + x_{3t} + x_{4t} = 50$$

(The number of goods received/shipped constraints)

$$x_{s1} - x_{13} - x_{1t} = 0$$

$$x_{s2} - x_{23} = 0$$

$$x_{s3} + x_{13} + x_{23} - x_{34} - x_{3t} = 0$$

$$x_{34} - x_{4t} = 0$$

(The continuity of goods constraints)

$$x_{ij} \leq u_{ij}$$

(The capacity constraints)

$$x_{ij} \in \mathbb{R}, \ \forall (i, j) \in S$$

In standard form, we can rewrite this as:

$$\text{minimize} \sum_{i,j \in S} c_{ij} x_{ij}$$

s.t.

$$x_{s1} + x_{s2} + x_{s3} = 50$$

$$x_{1t} + x_{3t} + x_{4t} = 50$$

(The number of goods received/shipped constraints)

$$x_{s1} - x_{13} - x_{1t} = 0$$

$$x_{s2} - x_{23} = 0$$

$$x_{s3} + x_{13} + x_{23} - x_{34} - x_{3t} = 0$$

$$x_{34} - x_{4t} = 0$$

(The continuity of goods constraints)

$$x_{ij} + s_{ij} = u_{ij}$$

(The capacity constraints)

$$x_{ij} = x_{ij}^{+} - x_{ij}^{-}$$

$$x_{ij}^{+}, x_{ij}^{-}, s_{ij} \geq 0, \ \forall (i,j) \in S$$